

Urban Growth Patterns Emerging from a *Physarum*-Solver Model Coupled with Population Dynamics

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1. Introduction

Cities are complex systems of heterogeneous components.

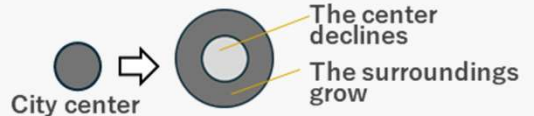


Important question:

“Can one simple mechanism generate multiple urban growth patterns?”



As a representative case, we focus on the **urban core hollowing**.

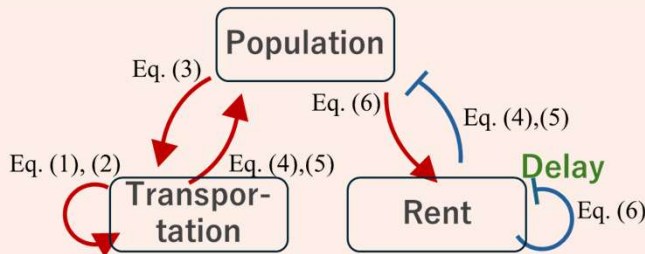


Purpose:

- 1: To simulate the urban core hollowing.
- 2: To test whether other multiple growth patterns can also emerge.

2. Hypothesis

We assume that **urban core hollowing** can emerge from three mechanisms.



1. **Positive feedback:**



2. **Negative feedback:**



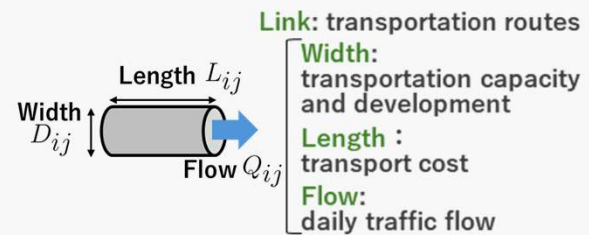
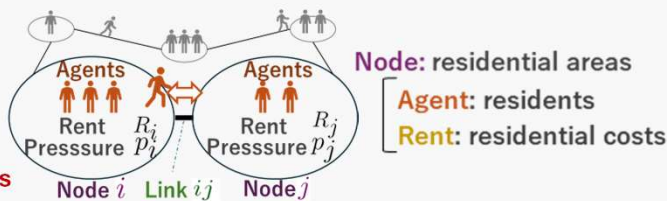
3. **Delayed rent adjustment:**



3. Model

Overview

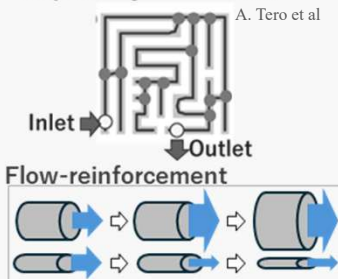
Coupling adaptive network and population dynamics



Update of Transportation Network

Physarum-solver model

A model of a tubular system inspired by Slime-mold



• Links with larger flows become wider

$$\dot{D}_{ij}(t) = f(|Q_{ij}(t)|) - D_{ij}(t) \quad \text{Eq.(1)}$$

$f(x)$: increasing function of x

• Developed links carry more flow

$$Q_{ij}(t) = \frac{D_{ij}(t)}{L_{ij}} (p_i(t) - p_j(t)) \quad \text{Eq.(2)}$$

• Larger populations generate larger traffic volumes (Our original)

$$I_0(t) = c_I (N_{inlet}(t) + N_{outlet}(t)) \quad \text{Eq.(3)}$$

Number of agents on the inlet node

Population Dynamics

• Agent can get birth or dead



• Agents prefer “lower rent”, “moderate population”, and “better transportation” nodes.

$$P_{k,i \rightarrow j}(t) = \varepsilon_2 \max(h_j(t) - h_i(t), 0) \quad \text{Eq.(4)}$$

$$h_i(t) = \frac{D_i(t)}{R_i(t) \cdot (1 + N_i(t))} \quad \text{Eq.(5)}$$

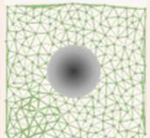
• Population increases rent value

$$\tau_R \dot{R}_i(t) = N_i(t) - R_i(t) + \varepsilon_R \quad \text{Eq.(6)}$$

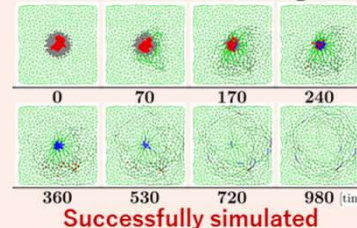
4. Simulation

Urban Core Hollowing

Initial condition



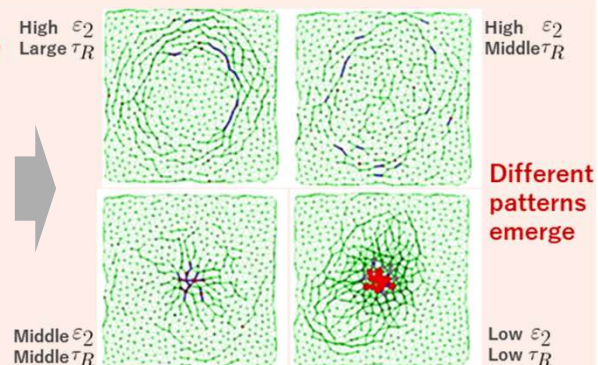
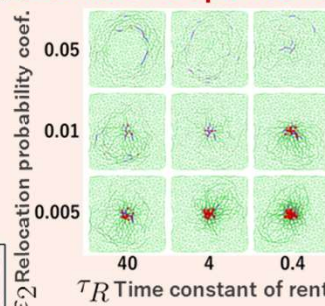
Gaussian distribution:
• Agents
• Line width
• Rent



Successfully simulated

Size: Num of agents (red dot: Top 10 nodes)
Width: Line width (blue line: Top 10 edges)

Parameter Dependence



Different patterns emerge

Conclusion:

A simple population-transport feedback model generates multiple urban-growth regimes.

Future work:

To test whether the same feedback framework explains other self-organizing spatial systems.

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